

Reliance on a classifier across three accuracy levels: the Tejada et al. (2022) sequential data

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Why this note

Liu, Zhou, Shen, Liu, Wu and Chen (CHI 2026) identify behavioral indicators of overreliance from interaction logs: copy-paste frequency, early task comprehension, window switching, coarse locating, and hesitation before adopting the model's output. The Tejada et al. (2022) data have no interaction logs, so those indicators cannot be computed on them. What the Tejada sequential design does allow is the trial-level reliance measures of the reliance literature (switch rate; RAIR and RSR from Schemmer et al. 2023; over- and underreliance), split by whether the classifier was right and by classifier accuracy level. Tejada et al. report only the overall switch proportion per classifier. A search of the 54 works citing Tejada et al. and of Steyvers' publications found no analysis with this split. This note reports it.

Data

Tejada et al. (2022), sequential paradigm: 72 participants after the exclusions used in my earlier reanalysis (OSF data, used with Mark Steyvers' permission), 24 per classifier level, 192 trials each, 13,824 initial/final pairs. Each trial: the participant classifies a noisy ImageNet image into one of 16 classes and rates confidence (low, medium, high), sees a VGG-19 classifier's output, then answers again. Between participants, the classifier is one of three fine-tuned versions. On these trials their accuracy is 45% (A, below the participants), 63% (B, about the participants' level) and 68% (C, above). All measures below are on disagreement trials, where the initial answer differs from the classifier's class (56% of trials). Intervals are 95% participant-cluster bootstrap, 2,000 draws.

Results

Measure (disagreement trials)	A (45%)	B (63%)	C (68%)
Switch rate: final answer = classifier	0.28 [0.23, 0.33]	0.50 [0.42, 0.57]	0.53 [0.44, 0.62]
RAIR: initial wrong, classifier right, switched	0.49 [0.43, 0.56]	0.62 [0.55, 0.69]	0.64 [0.51, 0.78]
RSR: initial right, classifier wrong, kept own	0.88 [0.84, 0.92]	0.69 [0.62, 0.75]	0.67 [0.60, 0.75]
Overreliance: initial right, classifier wrong, switched	0.07 [0.05, 0.10]	0.21 [0.15, 0.27]	0.22 [0.17, 0.27]
Underreliance: initial wrong, classifier right, not switched	0.51 [0.44, 0.57]	0.38 [0.31, 0.45]	0.36 [0.22, 0.49]
Final-answer accuracy on these trials	0.37 [0.34, 0.39]	0.33 [0.31, 0.35]	0.41 [0.36, 0.46]

Three things stand out.

1. The change is a step, not a slope. Between A and B every reliance measure moves, with intervals excluding zero (switch rate +0.22 [0.13, 0.30]; RSR -0.19 [-0.26, -0.12]; overreliance +0.14 [0.08, 0.21]). Between B and C no measure moves (every interval includes zero). Participants responded to the classifier going from below their own level to about their level; they did not respond further to it going above.

2. The cost of that step is on the trials where the person was right. Overreliance triples, from 7% to 21%: with a classifier at their own level, participants gave up a correct answer for a wrong one three times as often. RSR falls from 0.88 to 0.69. The gain, RAIR rising from 0.49 to 0.62, is smaller than the loss on this measure.

3. Underreliance stays high at every level. Even with the best classifier, participants failed to adopt a correct classifier answer on 36% of the trials where it would have corrected them.

The confidence breakdown (an analog of the CHI 'hesitation' pattern, not the same indicator) shows where the step lands: on trials where the participant was highly confident and the classifier was wrong, the switch rate is 9% under A, 35% under B, 26% under C. High initial confidence protected against a classifier that was visibly worse; it protected much less against one that was about as good. By final-stage response time, fast decisions against a wrong classifier switched 11% under A and 39% under B.

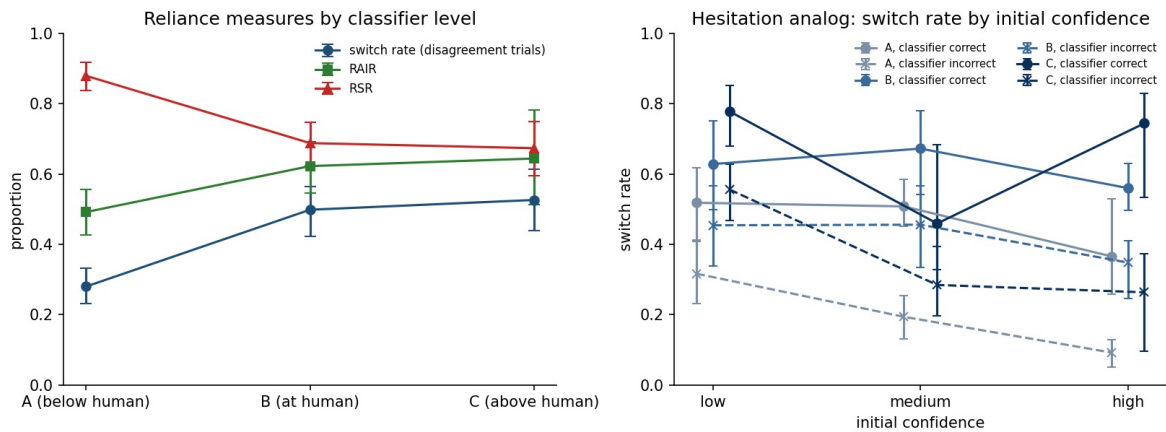


Figure. Left: switch rate, RAIR and RSR by classifier level with 95% intervals. Right: switch rate by initial confidence; solid = classifier correct, dashed = classifier incorrect.

Limits

One task, one participant pool, three classifier levels spanning 45% to 68%; levels are between participants, so the step could partly reflect who was assigned to A. Rates pool trials across participants within a level. The comparison with the CHI indicators is by construction only: no claim about those indicators follows from this note. Everything here was specified and committed before computation; the specification, its reading rules, the code and the full report are in the repository.